

24-S1-Q4

Q: HAC

$$Z_1 = [1.795 \quad 3.490 \quad 2.417]^T$$

$$Z_2 = [1.876 \quad 3.409 \quad 1.672]^T$$

$$Z_3 = [0.343 \quad 2.769 \quad 3.035]^T$$

$$Z_4 = [3.578 \quad -1.350 \quad 0.725]^T$$

(a) Euclidean + single  $\rightarrow$  process + dendrogram

(b) E + complete

(c) Silhouette + rationale

Solution (a) ① compute all pair wise distance among  $\{z_1, z_2, z_3, z_4\}$ .

$$\text{Denote } d_{ij} = \sqrt{\sum_{k=1}^3 (x_i - x_j)^2}$$

$$\begin{aligned} d_{12} &= \sqrt{(1.795 - 1.876)^2 + (3.490 - 3.409)^2 + (2.417 - 1.672)^2} \\ &= 0.7538 \end{aligned}$$

$$\begin{aligned} d_{13} &= \sqrt{(1.795 - 0.343)^2 + (3.490 - 2.769)^2 + (2.417 - 3.035)^2} \\ &= 1.7350 \end{aligned}$$

$$\begin{aligned} d_{14} &= \sqrt{(1.795 - 3.578)^2 + (3.490 + 1.350)^2 + (2.417 - 0.725)^2} \\ &= 5.4284 \end{aligned}$$

$$\begin{aligned} d_{23} &= \sqrt{(1.876 - 0.343)^2 + (3.409 - 2.769)^2 + (1.672 - 3.035)^2} \\ &= 2.1488 \end{aligned}$$

$$\begin{aligned} d_{24} &= \sqrt{(1.876 - 3.578)^2 + (3.409 + 1.350)^2 + (1.672 - 0.725)^2} \\ &= 5.1421 \end{aligned}$$

$$\begin{aligned} d_{34} &= \sqrt{(0.343 - 3.578)^2 + (2.769 + 1.35)^2 + (3.035 - 0.725)^2} \\ &= 5.7243 \end{aligned}$$

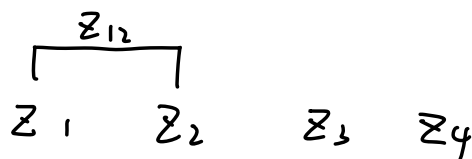
So the similarity matrix is as follows

|       | $z_1$  | $z_2$  | $z_3$  | $z_4$ |
|-------|--------|--------|--------|-------|
| $z_1$ | 0      |        |        |       |
| $z_2$ | 0.7538 | 0      |        |       |
| $z_3$ | 1.7350 | 2.1488 | 0      |       |
| $z_4$ | 5.4284 | 5.1421 | 5.7293 | 0     |

Obviously,  $z_1$  and  $z_2$  have the minimum distance 0.7538 and are clustered together.

The new cluster is named as  $z_{12}$ .

The dendrogram is as follows



② Next, we compute the similarity matrix of the  $N-1$  clusters.

$$\begin{aligned}
 d_{z_{12}, z_3} &= \min \{ d_{1,3}, d_{2,3} \} \\
 &= \min \{ 1.7350, 2.1488 \}
 \end{aligned}$$

$$= 1.7350$$

$$d_{z_{12}, z_4} = \min \{d_{1,4}, d_{2,4}\}$$

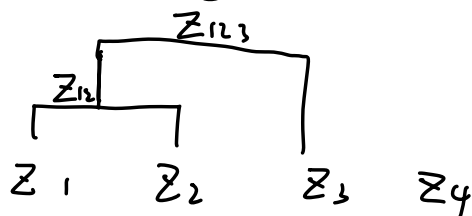
$$= 5.1421$$

|          | $z_{12}$ | $z_3$  | $z_4$ |
|----------|----------|--------|-------|
| $z_{12}$ | 0        |        |       |
| $z_3$    | 1.7350   | 0      |       |
| $z_4$    | 5.1421   | 5.7243 | 0     |

Obviously,  $z_{12}$  and  $z_3$  have the minimum distance 1.7350 and are clustered together

The new cluster is named as  $z_{123}$

The dendrogram is as follow



$$\textcircled{3} \quad d_{z_{123}, z_4} = \min \{ d_{1,4}, d_{2,4}, d_{3,4} \}$$

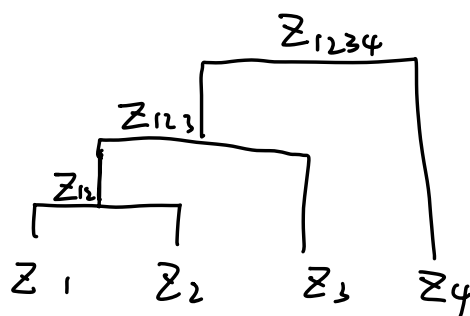
$$= 5.1421$$

|           | $z_{123}$ | $z_4$ |
|-----------|-----------|-------|
| $z_{123}$ | 0         |       |
| $z_4$     | 5.1421    | 0     |

Obviously,  $z_{123}$  and  $z_4$  have the minimum distance 5.1421 and are clustered together

The new cluster is named as  $z_{1234}$

The dendrogram is as follow



(b) ① compute all pair wise distance among  $\{z_1, z_2, z_3, z_4\}$ .

$$\text{Denote } d_{ij} = \sqrt{\sum_{k=1}^3 (x_i - x_j)^2}$$

$$\begin{aligned} d_{12} &= \sqrt{(1.795 - 1.876)^2 + (3.490 - 3.409)^2 + (2.417 - 1.672)^2} \\ &= 0.7538 \end{aligned}$$

$$\begin{aligned} d_{13} &= \sqrt{(1.795 - 0.343)^2 + (3.490 - 2.769)^2 + (2.417 - 3.035)^2} \\ &= 1.7350 \end{aligned}$$

$$\begin{aligned} d_{14} &= \sqrt{(1.795 - 3.578)^2 + (3.490 + 1.350)^2 + (2.417 - 0.725)^2} \\ &= 5.4284 \end{aligned}$$

$$\begin{aligned} d_{23} &= \sqrt{(1.876 - 0.343)^2 + (3.409 - 2.769)^2 + (1.672 - 3.035)^2} \\ &= 2.1488 \end{aligned}$$

$$\begin{aligned} d_{24} &= \sqrt{(1.876 - 3.578)^2 + (3.409 + 1.350)^2 + (1.672 - 0.725)^2} \\ &= 5.1421 \end{aligned}$$

$$\begin{aligned} d_{34} &= \sqrt{(0.343 - 3.578)^2 + (2.769 + 1.35)^2 + (3.035 - 0.725)^2} \\ &= 5.7243 \end{aligned}$$

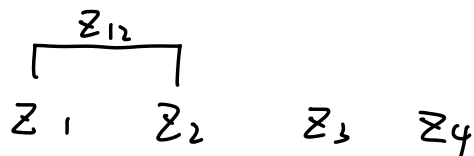
So the similarity matrix is as follows

|       | $z_1$  | $z_2$  | $z_3$  | $z_4$ |
|-------|--------|--------|--------|-------|
| $z_1$ | 0      |        |        |       |
| $z_2$ | 0.7538 | 0      |        |       |
| $z_3$ | 1.7350 | 2.1488 | 0      |       |
| $z_4$ | 5.4284 | 5.1421 | 5.7243 | 0     |

Obviously,  $z_1$  and  $z_2$  have the minimum distance 0.7538 and are clustered together

The new cluster is named as  $z_{12}$

The dendrogram is as follow



② Next, we compute the similarity matrix of the  $N-1$  clusters.

$$\begin{aligned}
 d_{z_{12}, z_3} &= \max \{ d_{1,3}, d_{2,3} \} \\
 &= \max \{ 1.7350, 2.1488 \}
 \end{aligned}$$

$$= 2.1488$$

$$d_{z_{12}, z_4} = \max \{d_{1,4}, d_{2,4}\}$$

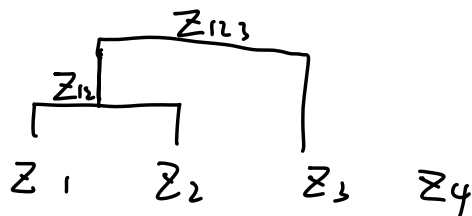
$$= 5.4282$$

|          | $z_{12}$ | $z_3$  | $z_4$ |
|----------|----------|--------|-------|
| $z_{12}$ | 0        |        |       |
| $z_3$    | 2.1488   | 0      |       |
| $z_4$    | 5.4282   | 5.7243 | 0     |

Obviously,  $z_{12}$  and  $z_3$  have the minimum distance 2.1488 and are clustered together

The new cluster is named as  $z_{123}$

The dendrogram is as follow





$$\textcircled{3} \quad d_{z_{123}, z_4} = \max \{ d_{1,4}, d_{2,4}, d_{3,4} \}$$

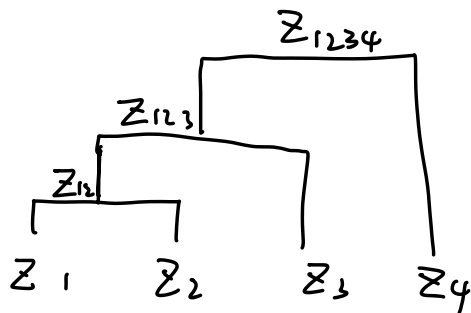
$$= 5.7243$$

|           | $z_{123}$ | $z_4$ |
|-----------|-----------|-------|
| $z_{123}$ | 0         |       |
| $z_4$     | 5.7243    | 0     |

Obviously,  $z_{123}$  and  $z_4$  have the minimum distance 5.7243 and are clustered together

The new cluster is named as  $z_{1234}$

The dendrogram is as follow



(C) The Silhouette coefficient for a particular sample  $x_i$  is defined as follows

$$SC_i = \frac{b_i - a_i}{\max\{a_i, b_i\}}$$

where  $a_i$  is the average distance between  $x_i$  and all the other samples within the same cluster

$$a_i = \frac{1}{n_p - 1} \sum_{x_i, x_j \in D_p, i \neq j} d(i, j)$$

where  $d(i, j)$  denotes the distance between  $x_i$  and  $x_j$

$b_i$  is the minimum distance between  $x_i$  and all other samples belonging to other clusters

$$b_i = \min_{x_i \in D_p, x_j \in D_q, p \neq q} d(i, j)$$

The overall Silhouette coefficient is

obtained as follows:

$$SC = \frac{1}{N} \sum_{i=1}^N SC_i$$

the ranges from  $-1$  to  $1$

A higher Silhouette coefficient indicates that the samples are well-clustered with clear separation between clusters and tight cohesion within each cluster

Conversely, a lower Silhouette coefficient suggests that the clustering may be less accurate, with overlapping clusters or sample that are not well-assigned to their respective cluster

## ② Rationale

The Silhouette coefficient is a metric that measures how well each sample fits

into its assigned cluster.. It combines information about both

(1) Cohesion : how close a sample is to other samples in its own cluster

(2) Separation : how far a sample is from samples in other clusters.